Stephen J. M. McKay

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Summary

I am interested in massive galaxy formation and evolution at early times, particularly the dust-obscured phase known as dusty star-forming galaxies (DSFGs), the most rapidly star-forming galaxies in the Universe. Using state-of-the-art telescopes such as the Atacama Large Millimeter/submillimeter Array (ALMA) and the James Webb Space Telescope (JWST) I study the stellar and dust properties and morphologies of these galaxies to help understand their role in the history of cosmic star formation and structure assembly.

I am also interested in statistical and machine learning methods and their application in a wide range of scientific problems.

I currently work with Prof. Amy Barger at the University of Wisconsin–Madison and recently defended my PhD dissertation.

Education

2026	**PhD, Physics**, University of Wisconsin–Madison GPA: 3.87 Dissertation: "The Physical Properties and Redshifts of Dusty Star-forming Galaxies: Observational Studies with ALMA and JWST" Advisor: Prof. Amy Barger, Dept. of Astronomy
2023	**MA, Physics**, University of Wisconsin–Madison GPA: 3.90
2021	**BS, Physics and Mathematics**, Wheaton College (IL) GPA: 3.99, *summa cum laude* Honors Thesis: "Implementation of Angular Momentum Transport by an Accretion Disk in MESA" Advisor: Dr. A. J. Poelarends

Publications (ADS library)

First Author

- **McKay, S. J.**; Barger, A. J; Cowie, L. L.; & Bauer, F. E. (2026). "A Deep ALMA Survey of the Redshift Distribution of Dusty Star-forming Galaxies." arXiv:2604.11879. ApJ, *accepted*
- **McKay, S. J.**; Barger, A. J; Cowie, L. L.; & Nicandro Rosenthal, M. J. (2025). "The Physical Properties and Morphologies of Faint Dusty Star-forming Galaxies Identified with JWST." ApJ, 988, 135.
- **McKay, S. J.**; Barger, A. J; & Cowie, L. L. (2024). "Comparing SCUBA-2 and ALMA Selections of Faint Dusty Star-forming Galaxies in A2744." ApJ, 962, 128.
- **McKay, S. J.**; Barger, A. J; Cowie, L. L.; Bauer, F. E.; & Nicandro Rosenthal, M. J. (2023). "Dust Properties of 870 μm-selected Galaxies in GOODS-S." ApJ, 951, 48.

Co-Author

- Barger, A. J; Cowie, L. L.; **McKay, S. J.**; & Bauer, F. E. (2026). "A Redshift-based Red Selection of Dusty Star-forming Galaxies." ApJ, 1004, 241.
- Nicandro Rosenthal, M. J., **McKay, S. J.**; Barger, A. J; & Cowie, L. L. (2026). "Molecular Gas Detections in Eight Faint DSFGs with Red NIR Colors at $z = 1.2$–2.5." arXiv:2601.20862. ApJ, *submitted*
- Nicandro Rosenthal, M. J.; Barger, A. J; Cowie, L. L.; Jones, L. H.; **McKay, S. J.**; & Taylor, A. J. (2025). "Spectroscopic Confirmation of a Massive Protocluster with Two Substructures at $z \simeq 3.1$." ApJ, 979, 247.

White Papers

- Polzin, A., Whitaker, K. E., Urry, C. M., et al. (2025). Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics. arXiv:2512.24465

Research Talks

2026	"New Views of Dusty Star-forming Galaxies with JWST and ALMA." *247th Meeting of the American Astronomical Society*. Phoenix, AZ.
2025	"Spectroscopic Redshifts for Faint Dusty Star-forming Galaxies." *2025 Wisconsin Space Conference, University of Wisconsin–Green Bay.* Green Bay, WI.
2025	"A New View of Dusty Star-forming Galaxies with JWST and ALMA." *New Data that Challenge Underlying Assumptions in Early Galaxy Evolution, CHOIR collaboration workshop, Schoodic Institute.* Winter Harbor, ME.
2025	"Revealing Faint Dusty Star-forming Galaxies with JWST and ALMA." *Extragalactic Discussion Group Seminar, University of Hawaii Institute for Astronomy.* Honolulu, HI.
2025	"Revealing Faint Dusty Star-forming Galaxies with JWST and ALMA." *Instituto de Astrofísica, Pontificia Universidad Católica*. Santiago, Chile.
2024	"The Physical Properties of Faint Dusty Star-forming Galaxies in GOODS-S and A2744." *244th Meeting of the American Astronomical Society*. Madison, WI.

Poster Presentations

2024	"Identifying Faint Dusty Star-forming Galaxies with JWST NIRCam." ESA/STScI conference: *Science with the Hubble and James Webb Space Telescopes VII: Stars, Gas, and Dust in the Universe*, Porto, PT.
2024	"Comparing ALMA and SCUBA-2 Selections of DSFGs in Abell 2744." ALMA Ambassadors Poster Session, National Radio Astronomy Observatory, Charlottesville, VA.
2023	"Unveiling the DSFG Population with a Red NIRCam Selection." STScI *First Year of JWST Science* Conference. Space Telescope Science Institute, Baltimore, MD.
2019	"Emission and Current Density Distribution in an Extended Magnetic Arcade." Society of Physics Students *Physics Congress 2019*. Providence, RI.

Conference Sessions

2025	Co-led *"Growing and Destroying Dust"* breakout session at CHOIR collaboration workshop, focused on understanding the existing paradigms and major open questions in dust formation and evolution.

Selected Awards & Honors

2024	Graduate Research Fellowship, Wisconsin Space Grant Consortium
2024	ALMA Ambassador Fellow, National Radio Astronomy Observatory
2022	Best Teaching Assistant Spring 2022, Dept. of Physics, UW–Madison
2020	Barry M. Goldwater Scholarship
2020	Induction into Sigma Pi Sigma Honors Society
2020	Joseph Spradley Outstanding Physics Student, Dept. of Physics, Wheaton College
2020	Senior Scholarship (1 of 6), Wheaton College Alumni Association
2019 & 2020	Physics Merit Scholarship (1 of 4), Wheaton College

Observing Experience

2025	Keck II 10-m telescope – KCWI – 2 nights integral field unit spectroscopy
2023–2025	Keck II 10-m telescope – DEIMOS – 5 nights multi-object spectroscopy
2023–2025	Keck I 10-m telescope – MOSFIRE – 7 nights multi-object spectroscopy

Teaching Experience

2021–2022 Teaching Assistant, *Dept. of Physics, UW–Madison*
Course: Physics 103 – Mechanics
6 discussion sections and 3 labs weekly, with 75 undergraduate students per semester.
Named "Best TA" based on aggregate student reviews.

2019–2021 Observatory Assistant, *Wheaton College*
Course: Astronomy 305
Operated two deck telescopes and one 24 in dome telescope, 3 hours weekly.
45 undergraduate students per semester.

Outreach and Volunteering

2024–present Activities for Community Outreach in STEM (ACORNS), *UW–Madison*
Leading astronomy & physics after-school educational nights (1-2 per month) for K-12 students through partnerships with local community centers, developing ongoing relationships and offering hands-on exposure to STEM activities.

2024 ALMA Ambassador Fellow, *NRAO/UW–Madison*
Supported new and current ALMA users by training them in radio astronomy methods, supporting community workshops, and engaging in science outreach and publicity. Planned and led all-day workshop, including presenting five 30–60 minute technical talks on proposal design and observing techniques and facilitating hands-on proposal design session.

2022	PEOPLE Program, *UW–Madison* Helped teach a short-term physics summer class for high-school students from underrepresented minority groups.

Training and Workshops

2025	CHOIR conference: New Data that Challenge Underlying Assumptions in Early Galaxy Evolution
2025	Picture an Astronomer Symposium
2024	Code/Astro: A Software Engineering Workshop for Astronomy
2023	IMPRS (Max Planck) Summer School: Galaxy Evolution with JWST
2023	SMA Interferometry School
2022	Penn State Summer School for Statistics for Astronomers XVII
2022	NRAO 18th Synthesis Imaging Workshop

Additional Projects

2026 Galaxy Morphology Classification

- Trained a convolutional neural network (PyTorch) to classify galaxy morphologies using public CANDELS imaging datasets of 10K–100K galaxies, applying data transforms to take advantage of inductive biases in training.
- Testing model against more traditional ML methods (e.g., PCA + random forest) and designing a visualization module to cleanly display results.

2024 Simulated Galaxy Inference, *Code/Astro Software Engineering Workshop, UChicago*

- Trained a random forest regression model on features from simulated galaxies to infer key galaxy properties, achieving an $\sim$80% accuracy score on testing data.

Skills

Programming:	Python (NumPy, SciPy, Pandas, PyTorch, Scikit-learn, PyMC, Emcee, Matplotlib, Astropy), Bash (Linux), MATLAB, Java, C/C++
Modeling & Data:	Regression, Bayesian inference (e.g., Markov chain Monte Carlo), signal detection, spectral energy distribution fitting, image analysis, uncertainty estimation, pipeline design, radio interferometry
Other software & tools:	Git, VS Code, Jupyter, Google Colab, CASA, CARTA
Observing/lab skills:	Optical/near-IR spectroscopy, CCD imaging, Oscilloscopes, Soldering
Languages:	Spanish (limited working proficiency)